Sruthi Rachamalla

Ph.D. in Computer Science | Software Engineer | Dallas, Texas

sruthirachamalla@siu.edu | Google Scholar | ORCID: 0000-0001-9312-507X

RESEARCH FOCUS

Artificial intelligence and intelligent systems, with a focus on behavioral modeling, cooperative decision-making, and incentives for safer human actions. Research connects intelligent transportation and blockchain-based driver rewards with LLM orchestration, cloud systems, and enterprise interoperability.

EDUCATION

Ph.D., Computer Science December 2023
Southern Illinois University Carbondale

Dissertation: A Framework for Crypto-Based Monetization of Driver Behavior with Blockchain.
Advisor: Henry Hexmoor.

M.S., Computer Science May 2017
Southern Illinois University Carbondale

Thesis: Co-Operative Platooning Distributed Inter-Vehicle Argumentation.
Advisor: Henry Hexmoor.

PROFESSIONAL EXPERIENCE

Software Engineer April 2026 - present
VSSIQ Inc.

Develops AI applications using LLMs, retrieval-augmented generation, and machine learning. Builds Python and Java services, REST APIs, and distributed microservices; engineers cloud applications using AWS, Docker, Kubernetes, and CI/CD.

Software Engineer April 2024 - March 2026
Infoville Inc.

Developed Python pipelines for customer-data integration, entity resolution, and segmentation. Built an AWS-based recommendation engine integrating feature pipelines and LightGBM ranking with Lambda, S3, DynamoDB, and SageMaker.

HONORS

Best Paper Finalist Award, CATA 2022 - DSRBT - Driving Safety Reward based on Blockchain Technology.

Outstanding Graduate Teaching Assistant, Southern Illinois University Carbondale, 2019-2020.

TECHNICAL TOOLKIT

Python, Java, Bash; JavaScript, React, Django; PyTorch, TensorFlow, scikit-learn, NumPy, Pandas; SQL/MySQL; AWS (Lambda, S3, EC2, SageMaker, DynamoDB); Git, Docker, Kubernetes, and CI/CD.

RESEARCH RECORD / 01

Publications & scholarship

Published work, an openly available preprint, and the doctoral dissertation.

01. Human-Assured Agentic Interoperability for Resilient B2B Systems: A Deterministic Replay Benchmark for Procurement Message Transformation

V. R. Divi, S. Gullapalli, N. K. R. Muppidi, **S. Rachamalla**, and V. Bikkavolu.

ICERA 2026, pp. 481-486. DOI

02. GuardGRPC: Priority-Aware Resource Governance for OTLP/gRPC Telemetry Export

S. Gullapalli, V. R. Divi, N. K. R. Muppidi, **S. Rachamalla**, and S. Tatavarthi.

ICERA 2026, pp. 272-277. DOI

03. CostAgent: Self-Improving Autonomous LLM-Based Orchestration for Cost-Optimal Cloud Data Processing at Scale

N. K. R. Muppidi, V. R. Divi, S. Gulapalli, **S. Rachamalla**, S. Tatavarthi, and M. A. Lakshmi.

ICAISET 2026, pp. 1-6. DOI

04. Blockchain and IoT-Enabled Earning Mechanism for Driver Safety Rewards in Cooperative Platooning Environment

E. Maghsoudlou, **S. Rachamalla**, and H. Hexmoor.

International Journal of Emerging Trends in Engineering Research, 13(2), 39-49 (2025). DOI

05. Improving Road Safety by Blockchain-based Monetization of Driver Behavior

S. Rachamalla and H. Hexmoor.

International Journal of Computers and Their Applications, 29(3), 129-137 (2022). Publisher PDF

06. DSRBT - Driving Safety Reward based on Blockchain Technology

S. Rachamalla and H. Hexmoor.

CATA 2022, EPiC Series in Computing, 82, 72-81. DOI

07. Driver Safety Reward with Cooperative Platooning using Blockchain

S. Rachamalla and H. Hexmoor.

Preprint, arXiv:2312.02164 (2023). arXiv

08. A Framework for Crypto-Based Monetization of Driver Behavior with Blockchain

S. Rachamalla.

Doctoral dissertation, Southern Illinois University Carbondale (2023). University record

RESEARCH RECORD / 02

Forthcoming work & research

Publication status reflects the October 2026 academic record.

ACCEPTED PAPERS

09. When Should Cooperative Driving Earn a Reward? A Counterfactual Simulation Study

S. Rachamalla and H. Hexmoor.

Accepted, International Conference on Electrical and Computer Engineering Researches (ICECER 2026).

10. Closed-Loop DSRBT-AV: A Context-Aware Blockchain Reward Framework for Cooperative Autonomous Driving

S. Rachamalla and H. Hexmoor.

Accepted, Conference on Information Communications Technology and Society (ICTAS 2026).

11. CanvasAgent: A Context-Grounded LLM Framework for Dynamic Dashboard Composition in Enterprise CRM

H. K. Thirunagari, **S. Rachamalla**, B. K. Thota, S. Gullapalli, and V. R. Divi.

Accepted, International Conference on Artificial Intelligence Robotics, Signal and Image Processing (AIRoSIP 2026).

SUBMITTED MANUSCRIPT

12. A Comprehensive Survey of Intelligent Cyber Argumentation with Social Networking: From Research Foundations to System Development

B. K. Thota, X. Liu, and **S. Rachamalla**.

Submitted to Information (MDPI), 2026.

RESEARCH EXPERIENCE

Graduate Research Assistant - iFERM Fall 2022

Illinois Small Business Development Center, SIU Research Park

Analyzed cold-chain infrastructure and access to perishable-goods storage. Developed a Python and JavaScript web application to help stakeholders explore infrastructure data and identify regional gaps.

Graduate Research Assistant - ProMuViz Spring 2017

Computer Science, Southern Illinois University Carbondale

Developed an interactive protein-mutation visualization tool with filtering, zooming, and tooltips. Used Python and Plotly for heatmaps and 3D visualizations, with a responsive JavaScript interface and structured-data integration.

ACADEMIC COMMUNITY

Teaching, mentorship & service

Southern Illinois University Carbondale and professional communities.

TEACHING & MENTORING

Graduate Assistant - Extended Reality Development Academy Summer 2021 - Summer 2022
Illinois Small Business Development Center, SIU Research Park

Taught computer graphics and VR development; mentored 18 students across three cohorts. Guided capstone work including virtual building tours and a VR baseball game, and presented student projects at the research park.

Instructor Spring 2018 - Spring 2022
Computer Science, Southern Illinois University Carbondale

Taught foundational computing to undergraduate students across disciplines. Developed lectures, practical activities, assignments, and assessments, and incorporated student feedback into course delivery.

Teaching Assistant 2016; Fall 2017 - Spring 2018
Computer Science, Southern Illinois University Carbondale

Supported CS 200B (Computer Concepts), CS 201 (Problem Solving with Computers), and CS 455 (Advanced Data Structures & Algorithms). Led labs, graded coursework, and provided individual academic support.

Graduate Mentor Spring - Fall 2020
Computer Science, Southern Illinois University Carbondale

Mentored two graduate students on micro-cloud systems for cooperative and solar vehicles, supporting research design, data analysis, visualizations, reports, and presentations.

PROFESSIONAL & UNIVERSITY SERVICE

Peer review: IEEE Transactions on Intelligent Vehicles; IEEE Internet of Things Journal; Taylor & Francis journals; ACMSE, ICERA, ICUFN, WISE, and ICDAS. Sub-reviewer, CATA 2022.

Society of Women Engineers: Judge/reviewer and working-group member, 2026-2027. **Memberships:** IEEE, ACM, and SWE.

University service: ABET accreditation documentation, School of Computing (Summer 2021); international-student arrival support, Center for International Education (July 2018).

PRESENTATIONS

4G & 5G Industrial Revolution Technologies - Research Center in Latin America / Andean Parliament, virtual presentation (2022).

XR (Extended Reality) - with M. Brazley, Graduate School of Public Administration, Seoul National University, virtual presentation (2022).